formulas borrowed from mathematicians to quantify the risk-and-return relationship achieved through diversification.

Figure 3-1 illustrates the diversification concept that the Markowitz and other researchers talk about. Each stock in Figure 3-1 has two types of risk: market-based risk known as systematic or beta, and unique risk inherent in each stock that is also called unsystematic risk. The greater the number of stocks in a portfolio that are spread across all industry sectors, the lower the unique risk each stock has on the portfolio return. This leaves only beta or market risk. Markowitz argued that a total market portfolio is the lowest-risk portfolio because it is the most diversified.

Markowitz's research paper was submitted to and published in the prestigious *Journal of Finance*. Initially, his paper titled "Portfolio Selection" was little noticed. It was considered too basic by many academic authorities. The information was so intuitive that none of Markowitz's professors at the University of Chicago dreamed that this small report would change the way portfolios were viewed and managed at every level in finance.

Markowitz expanded his published research during 1959 in a book titled *Portfolio Selection: Efficient Diversification of Investments.* The work earned Markowitz wide recognition in the financial economics field and eventually earned him The Bank of Sweden Prize in Economic Sciences in Memory of Alfred Nobel. Most people refer to this award as a Nobel Prize even though that is not technically correct. There is no Nobel Prize in economics. There is only The Bank of Sweden Prize in Economic Sciences. Nonetheless, the idea revolutionized portfolio management.

FIGURE 3-1 Diversification Reduces Stock-Specific Risk